

5(a)(i)	mitral / bicuspid / atrioventricular, valve and left ventricle and semilunar / aortic, valve and aorta ;	1
5(a)(ii)	lack of, glucose / oxygen, to (some) brain cells (so cells cannot respire) ;	1
5(b)	any three from: 1 normally only SAN in (wall of) right atrium sends out impulses ; 2 ref. to extra impulses (from wall of left atrium) ; 3 atrial systole might occur when it should be in diastole ; 4 impulses interfere with each other / AW ; 5 SAN no longer able to act as a pacemaker ; 6 rapid since extra contractions ;	3

Question	Answer	Marks
5(c)(i)	X = carbonic anhydrase ; Y = haemoglobinic acid ;	2
5(c)(ii)	<i>idea of</i> a negative ion (chloride) entering RBC replaces a negative ion (hydrogencarbonate) leaving / reducing charge separation (across the membrane) / maintaining electrical neutrality of the cell ;	1
5(d)	tissue fluid ; I lymph	1
5(e)	<i>any three from:</i> 1 production of mucus by, mucous glands / goblet cells ; 2 mucus (is sticky and) traps particles of dust ; I pathogens, microbes, bacteria, etc. 3 cilia on ciliated epithelial cells ; A ciliated epithelium 4 cilia, waft / move, mucus / AW ; 5 <i>idea of</i> mucus with, particles / dust, is moved away from lung tissue ;	3